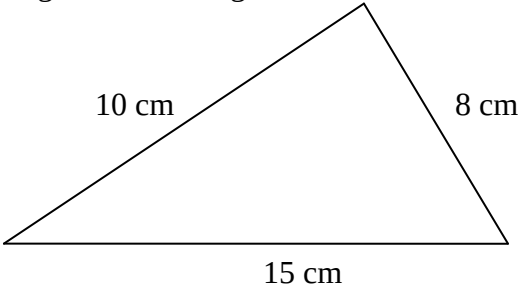
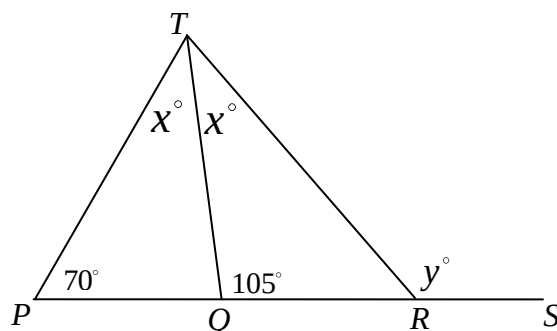


	BAULKHAM HILLS HIGH SCHOOL YEAR 11 MATHEMATICS TASK 2 JUNE 2009 <i>Time Allowed: 70 minutes</i>	
Instructions:	a) Write all your answers on your own paper. b) Show all necessary working. c) Marks may be deducted for careless or badly arranged work. d) No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.	
1. Find the exact value of; a) $\tan 330^\circ$ b) $\sec 120^\circ$		Marks 1 1
2. If $\tan \theta = \frac{3}{5}$ and θ is a reflex angle, find the exact value of $\cos \theta$		2
3. A ship leaves port and sails on a bearing of 120°T for 50 km. a) Draw a sketch to show this information. b) How far east of the port is it at this time, correct to the nearest km?		1 2
4. Sketch $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$		2
5. Find the largest angle in this triangle, correct to the nearest minute. 		3
6. Prove that the points $A(1,2)$, $B(-1,6)$ and $C(2,0)$ are collinear.		3
7. If $(2,3)$ is the midpoint of the two points $A(x,y)$ and $B(4,4)$, find the coordinates of A.		3
8. A line is represented by the equation $x + 2y - 6 = 0$. Find the equation of the line that is perpendicular to this line and passes through the point $(1,3)$.		3

Marks

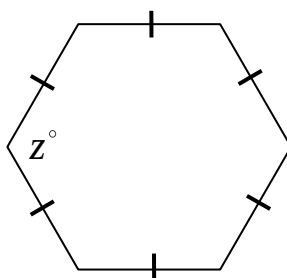
9. Find the value of each pronumeral, (no reasons required);

a)



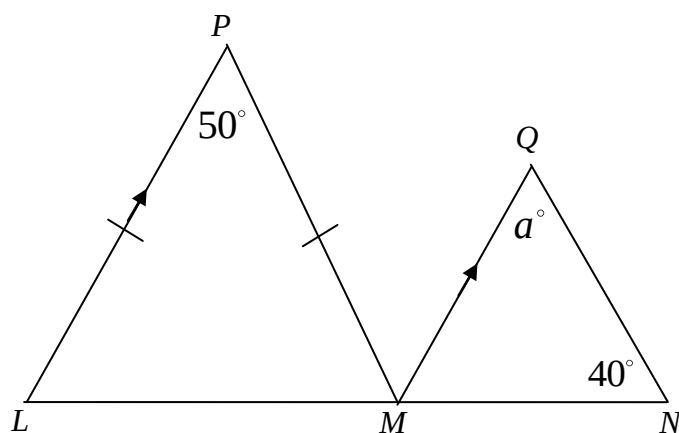
2

b)



1

c)



2

10. Solve for $0^\circ \leq \theta \leq 360^\circ$

a) $4\cos^2 \theta = 3$

3

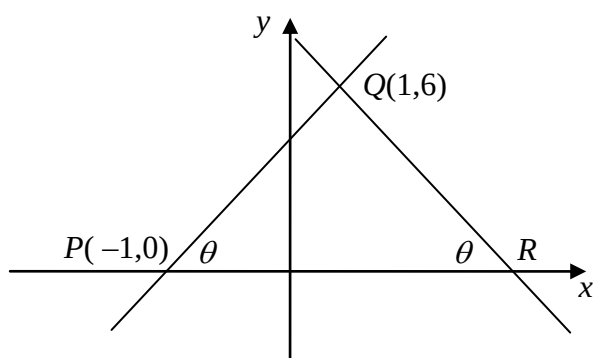
b) $2\sin \theta - 4\cos \theta = 0$

3

c) $\tan 2\theta = \frac{1}{\sqrt{3}}$

3

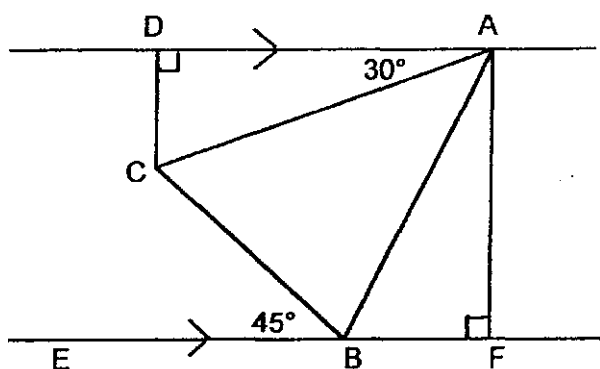
11.



In the diagram, P and Q have coordinates $(-1, 0)$ and $(1, 6)$ and $\angle QPR = \angle QRP = \theta$

- | | |
|---|---|
| a) Find the length of PQ as a surd. | 2 |
| b) Show that $\tan \theta = 3$ | 2 |
| c) Show that the gradient of QR is -3 . Give reasons. | 2 |
| d) Show that the equation of QR is $3x + y - 9 = 0$. | 2 |
| e) Find the coordinates of R . | 1 |
| f) Find the perpendicular distance from P to QR . | 2 |
| g) Hence calculate the area of $\triangle PQR$ | 2 |

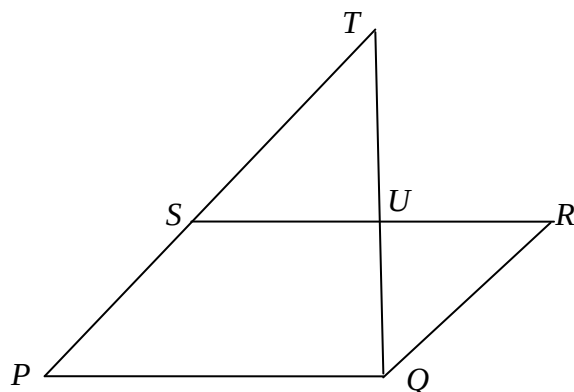
12.



In the diagram above $AC = AB$ and $DA \parallel EF$. $\angle DAC = 30^\circ$ and $\angle CBE = 45^\circ$. $CD \perp DA$ and $DA \perp AF$.

- | | |
|---|---|
| a) Find the size of $\angle ACB$, giving reasons. | 3 |
| b) Hence find the size of $\angle CAB$, giving reasons | 2 |
| c) Prove that $\triangle ACD \equiv \triangle ABF$ | 3 |

13.



$PQRS$ is a parallelogram and PT is a straight line through S . $TQ \perp PQ$

a) Prove $\triangle URQ \parallel \triangle QPT$

3

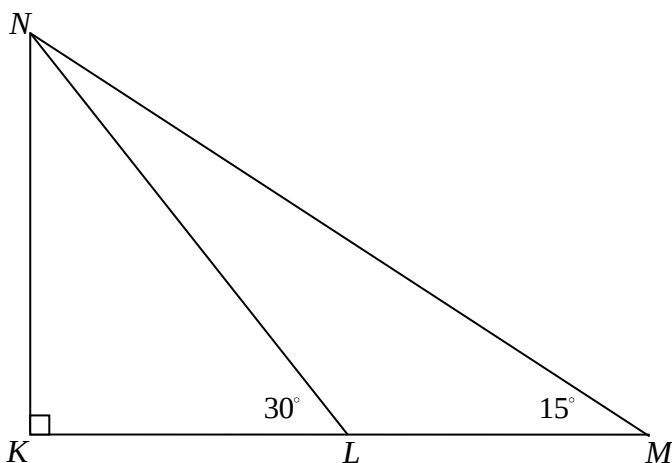
b) If $QR = 5$, $UR = 4$, $ST = 8$, find TP and SU and hence TQ .

5

14. Prove that $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta} = 2 \cot \theta$

2

15.



a) If $LM = 1$ metre, explain why LN also equals 1 metre.

2

b) Use the sketch to deduce that $\tan 15^\circ = 2 - \sqrt{3}$

3

Year 11 Mathematics Task 3 Solutions 1/471

1/a) $\tan 330^\circ = -\tan 30^\circ$
 $= -\frac{1}{\sqrt{3}}$ ①

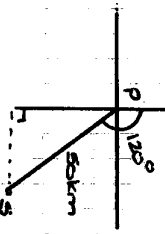
6/ $m_{AB} = \frac{6-2}{-1-1} = \frac{4}{-2} = -2$
 $m_{BC} = \frac{0-6}{2+1} = \frac{-6}{3} = -2$
 $\therefore m_{AB} = m_{BC}$
 $\therefore A, B, C$ are collinear ③

b) $\sec 120^\circ = \frac{1}{\cos 120^\circ}$
 $= \frac{1}{-\frac{1}{2}} = -2$ ①

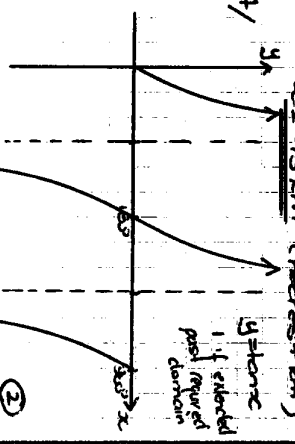
7/ $\frac{x+4}{2} = 2$
 $x+4 = 4$
 $x = 0$
 $y+4 = 6$
 $y = 2$
 $\therefore A$ is the point $(0, 2)$ ③

2/ $\tan \theta = \frac{3}{4}$ $\theta = \tan^{-1} \frac{3}{4}$
 $\therefore 103.7^\circ < \theta < 270^\circ$
 $\cos \theta = -\frac{4}{5}$ ②

8/ $x+2y-6 = 0 \Rightarrow m = -\frac{1}{2}$
 $\therefore$ required $m = 2$ ①

3/a)  ①

9/a) $x+70 = 105$
 $x = 35$ ①

4/  ③

b) $6z = 180(6-z)$
 $z = 120$ ①

5/ largest angle is opposite the longest side.
 $\cos \theta = \frac{8^2 + 10^2 - 15^2}{2 \times 8 \times 10} = -1$
 $\theta = 112.25^\circ$ ③

c) $2\angle PML + 50 = 180$
 $2\angle PML = 130$
 $\angle PML = 65^\circ$
 $\angle AMQ = 50^\circ$
 $a + 40 = 65 + 50$
 $a = 75^\circ$ ③

10/a) $4\cos^2 \theta = 3$
 $\cos^2 \theta = \frac{3}{4}$
 $\cos \theta = \pm \frac{\sqrt{3}}{2}$
 $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$ ③

b) $2\sin \theta - 4\cos \theta = 0$
 $2\sin \theta = 4\cos \theta$
 $\tan \theta = 2$ ①

11/a) $RQ = \sqrt{6^2 + 2^2} = \sqrt{40}$
 $= 2\sqrt{10}$ units ②

b) $m = \tan \theta = \frac{6-0}{-1-1} = -3$
 $\tan \theta = -\frac{3}{2}$ ②

c) $\tan 2\theta = \frac{1}{\sqrt{3}}$ $0^\circ < 2\theta < 720^\circ$
 $2\theta = 30^\circ, 210^\circ, 390^\circ, 570^\circ$
 $\theta = 15^\circ, 105^\circ, 195^\circ, 285^\circ$ ③

d) $\angle DAF + \angle BFA = 180^\circ$ (consecutive $\angle$ s $\parallel$ lines)
 $\angle BFA = 90^\circ$ (given)
 $\therefore \angle DAF = 90^\circ$
 $\angle DAF = \angle DAC + \angle CAB + \angle BAF$ (common $\angle$)
 $90 = 30 + 30 + \angle BAF$
 $\angle BAF = 30^\circ$
 $\angle BAF = \angle CAD$ (C.A.)
 $\angle AFB = \angle ADC = 90^\circ$ (given) (A)
 $AB = AC$ (given) (S)
 $\therefore \triangle ABF \cong \triangle ACD$ (AAS) ③

12/a) $\triangle ABC$ is isosceles
 $\therefore QR = 2\sqrt{10}$ ①

13/ $PT \parallel QR$ (opposite sides $\parallel$)
 $\angle PRT = \angle RQR$ (alternate $\angle$ s $\parallel$ lines)
 $\angle PRT = \angle RQR$ (given) (A)
 $\therefore \triangle PRT \cong \triangle RQR$ (AAS) ③

14/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

15/ $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

16/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

17/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

18/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

19/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

20/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

21/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

22/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

23/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

24/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

25/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

26/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

27/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

28/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

29/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

30/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

31/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

32/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

33/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

34/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

35/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

36/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

37/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

38/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

39/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

40/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

41/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

42/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

43/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

44/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

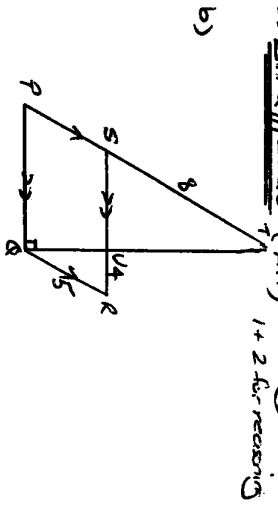
45/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

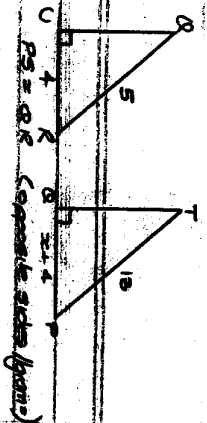
46/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

47/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

48/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③

49/a) $\triangle ABC$ is isosceles
 $\angle ABC = \angle ACB$
 $\angle ABC = 75^\circ$
 $\angle CAB + \angle ABC + \angle BAC = 180^\circ$ (sum of $\angle$ s)
 $\angle CAB = 30^\circ$ ③





$$PS = 5$$

$$PT = ST + PS = 13 - 1$$

$$\text{Let } SU = x,$$

$$RQ = RS \quad (\text{opposite sides of } \angle R = 90^\circ)$$

$$RQ = x + 4$$

$$\frac{x+4}{4} = \frac{13}{5} \quad (\text{ratio sides } \propto \text{Hypotenuse})$$

$$5x + 20 = 52$$

$$x = \frac{32}{5}$$

$$\therefore SU = \frac{32}{5} = 6.4$$

$$QT^2 = PT^2 - RQ^2$$

$$= 13^2 - \left(\frac{32}{5}\right)^2$$

$$= \frac{1681}{25} \quad \text{if } \angle R = 90^\circ$$

$$QT = \frac{41}{5} = 8.2$$

$$14/ \frac{1}{\cos \theta - \cos \theta} - \frac{1}{\cos \theta + \cos \theta}$$

$$= \frac{\cos \theta + \cos \theta}{\cos^2 \theta - \cos^2 \theta} - \frac{\cos \theta - \cos \theta}{\cos^2 \theta - \cos^2 \theta}$$

$$= \frac{2 \cos \theta}{0} = \infty$$

$$= \frac{2 \cos \theta}{0} = \infty$$

$$15/ \angle KLN = \angle LNM + \angle LNM$$

$$30 = \angle LNM + 15$$

$$\therefore \angle LNM = 15^\circ$$

$$\angle N = \angle M = 15^\circ$$

$$\angle N = \angle M = 15^\circ$$

$$b) \frac{KL}{1} = \cos 30^\circ$$

$$KL = \frac{\sqrt{3}}{2}$$

$$\frac{KN}{1} = \sin 30^\circ$$

$$KN = \frac{1}{2}$$

$$\tan 15^\circ = \frac{KN}{KL}$$

$$= \frac{KL}{KL + ML}$$

$$= \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2} + 1}$$

$$= \frac{1}{\sqrt{3} + 2} \times \frac{\sqrt{3} - 2}{\sqrt{3} - 2}$$

$$= \frac{\sqrt{3} - 2}{3 - 4}$$

$$= 2 - \sqrt{3}$$

$$(3)$$